Semester	7
Subject	DevOps Lab
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Assignment No	4
Assignment Title	To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat or nginx server.

Code	**File: src/main/java/com/vit/demo/Calculator.java**
	`package com.vit.demo;`
	
	`public class Calculator {`
	`  public int add(int a, int b) {`
	`  return a + b;`
	`  }`
	`}`
	
	File: src/test/java/com/vit/demo/CalculatorTest.java
	`package com.vit.demo;`
	
	`import org.junit.jupiter.api.Test;`
	`import static org.junit.jupiter.api.Assertions.assertEquals;`
	
	`public class CalculatorTest {`
	`  @Test`
	`  public void testAddition() {`
	`  Calculator calculator = new Calculator();`
	`  int result = calculator.add(10, 20);`
	`  assertEquals(30, result);`
	`  }`
	`}`
	
	File: src/main/webapp/index.jsp

```
<!DOCTYPE html>
<html>
<head>
    <title>Jenkins CI/CD Demo</title>
</head>
<body>
    <h1>Jenkins CI/CD Pipeline Successful!</h1>
    <h2>Application deployed on Apache Tomcat.</h2>
    <p>Build Tool: Maven</p>
    <p>CI/CD Tool: Jenkins</p>
    <p>Application Server: Apache Tomcat</p>
</body>
</html>
```

File: pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://maven.apache.org/POM/4.0.0
        https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.vit</groupId>
    <artifactId>jenkins-tomcat-demo</artifactId>
    <version>1.0</version>
            <!-- WAR packaging because the app deploys on Tomcat -->
    <packaging>war</packaging>

    <properties>
        <maven.compiler.release>21</maven.compiler.release>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>

    <dependencies>
        <!-- JUnit for testing -->
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>5.10.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <!-- Final WAR name -->
        <finalName>jenkins-demo</finalName>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.15.0</version>
            </plugin>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.5.4</version>
            </plugin>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-war-plugin</artifactId>
                <version>3.5.1</version>
            </plugin>
        </plugins>
    </build>
</project>
```

File: .gitignore

```
target/
.classpath
.project
.settings/
.idea/
.vscode/
```

File: Jenkinsfile

```
pipeline {
            agent any

            tools {
            jdk 'JDK-21'
            maven 'Maven-3'
            }

            stages {
        stage('Checkout') {
            steps {
                echo 'Downloading source code from GitHub'
                git branch: 'main',
                    url: 'https://github.com/YOUR-USERNAME/jenkins-tomcat-demo.git'
            }
            }

        stage('Compile') {
            steps {
                echo 'Compiling Java application'
                bat 'mvn clean compile'
            }
            }

        stage('Test') {
            steps {
                echo 'Executing JUnit tests'
                bat 'mvn test'
            }
            }

        stage('Package') {
            steps {
                echo 'Creating WAR file'
                bat 'mvn package -DskipTests'
                archiveArtifacts artifacts: 'target/*.war', fingerprint: true
            }
            }

        stage('Deploy') {
            steps {
                echo 'Deploying application to Tomcat'
                bat '''
                    copy /Y target\\jenkins-demo.war ^
                    "C:\\apache-tomcat-11.0.24\\webapps\\jenkins-demo.war"
                '''
            }
            }

        stage('Verify') {
            steps {
                echo 'Checking deployed application'
                bat '''
                    curl.exe --fail ^
                    http://localhost:8081/jenkins-demo/
                '''
            }
            }
            }

            post {
        always {
            junit allowEmptyResults: true,
                testResults: 'target/surefire-reports/*.xml'
```

```
            }
        success {
            echo 'CI/CD PIPELINE SUCCESSFUL'
            }
        failure {
            echo 'CI/CD PIPELINE FAILED'
            }
            }
}
```